

Machine learning and AI increasingly inform decisions across medicine, policy, and engineering. In these systems, reliable inference can face two complementary bottlenecks. When data are scarce or incomplete—as in many panel and tabular problems—the bottleneck is methodological: practitioners navigate a zoo of assumptions, estimators, and tuning choices, often spending weeks to reach an actionable answer. At the other extreme, when data are abundant or models are large—as in simulations and modern AI systems—the bottleneck shifts to computation, forcing users to discard data or accept less accurate estimates. My research addresses both: one thread turns complex panels into defensible counterfactuals without weeks of bespoke modeling; the other exploits structure to analyze more data within a fixed computational budget, without sacrificing statistical accuracy for the downstream task.

A unified framework for panel inference Imagine a social scientist deciding which intervention to deploy given panel data involving multiple units, staggered interventions, covariates, irregular missingness, and confounding. Even an expert may spend weeks choosing and tuning estimators and validating uncertainty, thereby delaying the decision. My goal is a theoretically supported, self-tuning method that returns reliable counterfactuals within hours—enough to compare interventions or design the next experiment—or flags insufficient evidence.

My group develops panel inference under weak nonparametric assumptions, emphasizing nearest-neighbor (NN) methods for their simplicity and interpretability. We built fine-grained NN counterfactual inference for personalized unit-by-time decisions in sequentially adaptive digital-health and social experiments;^[15] sharpened guarantees with doubly robust NN;^[14] and achieved minimax adaptation to unknown smoothness with a two-sided variant.^[24] A complementary line extended the framework beyond missing scalar outcomes to counterfactual forecasting,^[9] and distribution learning through kernel embeddings,^[8] Wasserstein barycenters,^[16] and exponential-family models.^[25]

Our goal is to replace today’s collection of problem-specific methods with a single, self-configuring framework. Within this framework, better matrix-completion algorithms directly sharpen doubly robust estimates, including for high-rank panels with unobserved confounding.^[1] Adaptively weighted NN removes tuning and cross-validation.^[23] TabImpute delivers state-of-the-art zero-shot imputation after roughly 12 hours on one GPU while inheriting advances in transformer training and synthetic-data generation,^[17] and GAAVI allows for anytime valid inference, opening a path to streaming decisions.^[7] Together, these components are making rigorous panel inference routine, freeing researchers to focus on scientific decisions rather than estimator design.

More inference per unit of computation Even after methodological choices are automated, the right method may be too expensive. Many methods—including NN for panels,^[15] kernel methods,^[11] and attention^[19]—scale quadratically with data or context length. Consequently, under fixed runtime or cost budgets, decision-makers may discard evidence, evaluate fewer alternatives, or accept less accurate estimates. My research develops structure-aware compression that makes many such computations nearly linear while retaining the precision of the full quadratic procedure—often reducing computational needs by 10–100× and turning weeks of computation into hours.

Kernel thinning (KT) began this program by compressing an n -point distribution into a near- $\sqrt{n}$ -point summary while preserving integration accuracy.^[13] Generalized KT extended these guarantees across kernels and distributions;^[12] Compress++ made compression near-linear in the input size;^[26] and Stein KT handled biased inputs such as non-converged Markov chains.^[22] Together, these methods enabled a *Compress-then-X* framework and provided quadratic speed-ups for hypothesis testing,^[11] kernel learning,^[21] counterfactual distribution learning,^[8] neural-network training,^[4] interpretability,^[3] experimental design,^[2] and mean-field dynamics.^[5]

Large language models (LLMs) are next: reasoning, long contexts, and agents increasingly require more test-time computation.^[10, 18] Thinformer, our noncausal precursor, cut dominant-layer runtime 5.6× while remaining within 0.4 percentage points of exact-attention ImageNet accuracy.^[4] Express makes the compression causal, achieving up to 82× faster 512K-token prefill and exact-attention accuracy with 61% of the cache or 56% of the decoding time.^[19] Our goal is a single statistical layer spanning inference, training, and agents that compounds with advances in hardware and systems, targeting an additional order-of-magnitude reduction in runtime or cost at the same fidelity.

From theory to measurable, reproducible progress N²-Bench,^[6] MissBench,^[17] and ORCA-bench^[20] measure validity and accuracy–cost–latency tradeoffs; open-source N², GoodPoints, TabImpute, Thinformer, and Express make the methods usable and extensible. This shared infrastructure accelerates cumulative progress toward autonomous decision systems limited only by the evidence—not by specialized expertise or prohibitive computation.

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